

Yisheng Tu

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EMPLOYMENT	Postdoctoral Research Fellow ▪ University of Michigan, Ann Arbor, MI, USA ▪ Supervisor: Prof. Lee Hartmann Lee Teng Internship, Fermi National Accelerator Laboratory ▪ Mentor: Dr. Tanaji Sen, Dr. Jean-Francois Ostiguy	Sep 2025 – Jun 2019 – Aug 2019
EDUCATION	Ph.D. in Astronomy ▪ University of Virginia, Charlottesville, VA, USA Bachelor of Science double major ▪ University of Rochester, Rochester, NY, USA ▪ Major: Physics and Astronomy & Mathematics ▪ Minor: Music	Sep 2020 – 2025 Sep 2016 – May 2020
PI PROPOSALS & FELLOWSHIPS	Discover ACCESS computational allocation (1.75M CPU-hour, \$7280) Interdisciplinary Fellowship (\$5000/yr), UVA	2025-2026 2020-2025
AWARDS	Chambliss Astronomy Achievement Award (AAS 244) Deans List, University of Rochester Research Presentation Awards, University of Rochester	2024 2016 – 2019 Mar 2019
TALKS AND PRESENTATIONS	Seminar Talk (CCA, New York City, NY) Astronomy Seminar (EPL, Washington, DC) 245th Americal Astronomy Society Meeting (National Harbor, MD) Seminar Talk (ASIAA, Taiwan, ROC) Journal Club (UM, Ann Arbor, MI) Star & Planet Formation Group (UNLV, Las Vegas, NV) Origins Seminar (UA, Tuscon, AZ) Monday Science Seminar (UW, Madison, WI) 244th Americal Astronomy Society Meeting (Madison, WI) Gordon Research Conference (South Hadley, MA) Athena ++ meeting (New York City, NY) Fermi National Accerelator Labortory (Batavia, IL) Death & Afterlives of stars (STScI, Baltimore, MD)	2025 2025 2025 2024 2024 2024 2024 2024 2024 2023 2023 2019 2019
REFeree	Astrophysical Journal, Astrophysical Journal Letter, Astronomy & Astrophysics	

STUDENT PROJECTS	Eric Rapp (University of Michigan)	2026 - current
	▪ Project: Heating due to interstellar object fly-through disks	
	Kass Bell (University of Virginia)	2025 - current
	▪ Project: MHD simulations of jets with a tilted stellar magnetosphere	
	Xiaoyuan Yang (Chinese University of Hong Kong)	2024 - current
	▪ Project: Synthetic observation of forming young protostellar systems	
	Samuel Crowe (University of Virginia)	2024 - 2025
	▪ Project: Radiation MHD Simulations of HII Regions near Galactic Center	
COURSE INSTRUCTED	Neha Bagalkot(University of Virginia)	2022
	▪ Project: Line Radiative Transfer in protostellar disk formation	
	Dylan Jones (University of Virginia)	2021 - 2022
	▪ Project: Centrifugal Barrier in Disk Formation	
	ASTR 1220: Introduction to stars, galaxies and the universe (UVA)	Summer 2023
OTHER WORK & VOLUNTEER EXPERIENCE	Public Night Coordinator, University of Virginia	2022-2023
	Teaching Assistant, University of Virginia	Spring 2021
	Student Summer Tour Guide, C. E. K. Mees Observatory	2018, 2019
	Teaching Intern, University of Rochester	Fall 2017 - Spring 2020
FIRST AUTHOR PUBLICATIONS	[1] Yisheng Tu , Xiaoyuan Yang, Zhi-Yun Li, “A New Method of Measuring Magnetic Field Strength in Highly Structured Protostellar Envelope”, in prep	
	[2] Yisheng Tu , Zhi-Yun Li, Zhaohuan Zhu, Kass Bell, “Modeling YSO Jets in 3D III: Dependency of accretion and jet outflow properties on stellar magnetic field strength and spin”, in prep	
	[3] Yisheng Tu , Zhi-Yun Li, Zhaohuan Zhu, Chun-Yen Hsu, Xiao Hu, “Modeling YSO Jets in 3D II: Accretion-Fed, Star-Anchored Poynting Jets in the Low-Density Polar Cavity Powered by Disk-Magnetosphere Interaction”, Accepted by ApJ	
	[4] Yisheng Tu , Zhi-Yun Li, Zhaohuan Zhu, Chun-Yen Hsu, Xiao Hu, “Modeling YSO Jets in 3D I: Highly Variable Asymmetric Magnetic Pressure-Driven Jets in the Polar Cavity from Toroidal Fields Generated by Inner Disk Accretion ”. In ApJ (July 2025), 988, 1, 107, 24.	
	[5] Yisheng Tu , Zhi-Yun Li, Zhaohuan Zhu, Xiao Hu, Chun-Yen Hsu, “YSO Jets Magnetocentrifugally Driven by Reconnecting Atmospheric Avalanche Accretion Streams Above Inner Circumstellar Disks”, ApJ, 978, 1, id.84, 13 pp. (Jan. 2025)	
	[6] Yisheng Tu , Zhi-Yun Li, Zhaohuan Zhu, Chun-Yen Hsu, “Fragmentation of Dense Rotation-Dominated Structures Fed by Collapsing Gravomagneto-Sheetlets and Origin of Misaligned 100 au-Scale Binaries and Multiple Systems”. In: mnras 532.3 (Aug. 2024), pp. 3135–3150	
	[7] Yisheng Tu , Zhi-Yun Li, Ka Ho Lam, Kengo Tomida, Chun-Yen Hsu, “Protostellar discs fed by dense collapsing gravo-magneto sheetlets”. In: mnras 527.4 (Feb. 2024), pp. 10131–10150	
	[8] Yisheng Tu , Zhi-Yun Li, Ka Ho Lam “Grain Growth During Protostellar Disk Formation”. In: mnras 515.4 (Oct. 2022), pp. 4780–4796.	

- [1] Samuel Crowe, **Yisheng Tu**, Zhi-Yun Li, “MHD Simulations of Strongly Magnetized HII Region Evolution: Evidence for Ionized Gas Filamentation”, submitted to ApJ
- [2] Chun-Yen Hsu, Zhi-Yun Li, **Yisheng Tu**, Xiao Hu, Min-Kai Lin, “Dust Concentration Via Coupled Vertical Settling and Radial Migration in Substructured Non-Ideal MHD Discs and Early Planet Formation ”, In: MNRAS 540, 1 (June 2025), pp 1363–1377
- [3] Chun-Yen Hsu, Zhi-Yun Li, **Yisheng Tu**, Xiao Hu; Min-Kai Lin, “Rossby Wave Instability and Substructure Formation in 3D Non-Ideal MHD Wind-Launching Disks”. In: ,mnras 533.3 (Sept. 2024), pp. 2980–2996
- [4] Renato Mazzei, Zhi-Yun Li, Che-Yu Chen, **Yisheng Tu**, Laura Fissel, Richard I. Klein, “Modelling CN Zeeman effect observations of the envelopes of a low-mass protostellar disc and a massive protostar”. In: mnras 527.3 (Jan. 2024), pp. 8618–8632.
- [5] Luke Chamandy, Jonathan Carroll-Nellenback, Eric G. Blackman, Adam Frank, **Yisheng Tu**, Baowei Liu, Yangyuxin Zou, Jason Nordhaus, “How negative feedback and the ambient environment limit the influence of recombination in common envelope evolution”. In: mnras 528.1 (Feb. 2024), pp. 234–254.
- [6] Dylan Jones, Ka Ho Lam, Zhi-Yun Li, **Yisheng Tu** “Centrifugal Barrier and Super-Keplerian Rotation in Protostellar Disk Formation”. In: mnras 517.1 (Nov. 2022), pp. 213–221.
- [7] Luke Chamandy, Eric G. Blackman, Adam Frank, Jonathan Carroll-Nellenback and **Yisheng Tu** “Common envelope evolution on the asymptotic giant branch: unbinding within a decade?”. In: mnras 495.4 (July 2020), pp. 4028–4039.
- [8] T. Sen, **Y. Tu**, J.-F. Ostiguy “Fields and Characteristic Impedances of Dipole and Quadrupole Cylindrical Stripline Kickers”. In: Physical Review Accelerators and Beams 23.1, 012801 (Jan. 2020), p. 012801.
- [9] Luke Chamandy, Eric G. Blackman, Adam Frank, Jonathan Carroll-Nellenback, Yangyuxin Zou and **Yisheng Tu** “How Drag Force Evolves in Global Common Envelope Simulations”. In: mnras 490.3 (Dec. 2019), pp. 3727–3739.
- [10] Luke Chamandy, **Yisheng Tu**, Eric G. Blackman, Jonathan Carroll-Nellenback, Adam Frank, Baowei Liu and Jason Nordhaus. “Energy Budget and Core-Envelope Motion in Common Envelope Evolution”. In: mnras 486.1 (June 2019), pp. 1070–1085.
- [11] Luke Chamandy, Adam Frank, Eric G. Blackman, Jonathan Carroll-Nellenback, Baowei Liu, **Yisheng Tu**, Jason Nordhaus, Zhuo Chen, Bo Peng. “Accretion in common envelope evolution”. In: mnras 480.2 (Oct. 2018), pp. 1898–1911.